Supplementary material

Reading the Moon's Plasma with a Neural Network: physics-informed inversion of Chandrayaan-3 RAMBHA-LP Langmuir probe sweeps

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Supplementary Appendix A. Claims withdrawn during this work

An earlier draft reported that this archive resolves two electron populations, with specific temperatures and a hot fraction. It also reported a recovery count that included sweeps clipped by the instrument. Those are withdrawn. They are listed here with reasons, because anyone who saw the earlier draft is entitled to know what changed.

Supplementary Table S1: *Claims made in an earlier draft, and the reason each was withdrawn.*

Withdrawn claim	Why it was withdrawn
The archive shows two electron populations	The supporting measurement had no control. Averaging single-population sweeps with a spread of temperatures also produces a graded signature.
Cold 2,856 K, hot 13,662 K, hot fraction 4.3%	Unstable. Three separate groupings of 1,498 sweeps each gave hot fractions from 0.1% to 5.3% - a factor of 50 - with identical cold density.
Two populations reconcile our temperature with the published value	Void, since the premise is unsupported.
Our result is in tension with the published analysis	There is no tension. With the control applied we agree with them.
The hot component is photoelectrons from the probe	Moot, and wrong in mechanism: probe photoemission is constant below the plasma potential and is already removed in the rawB data, with any remaining constant part taken out by the baseline subtraction, so it cannot produce the observed shape.
The network recovers 8,343 classical failures, raising usable coverage to 83.5%	2,935 of those sweeps were clipped by the instrument, and on clipped sweeps the network's values are artefacts: against the classical value on the unclipped twin half, density was 68% high and temperature 0.43 eV low. With clipped sweeps excluded the figures are 5,410 and 73.7%.
The network and the classical fit agree to about 1% in density	That is the median offset between them. Individual sweeps differ by 4.7% in density and 27% in temperature, and the report now gives both.

Supplementary Appendix B. How this work was done

This appendix exists because the method that produced the results above is, in our view, the most transferable part of the project.

Six times during this work we formed an explanation, tested it, and found it wrong. The hot population attributed to photoelectrons. The residual attributed to the sheath exponent - which, as one line of reasoning showed afterwards, cannot even act in the region where the signal appeared.

The residual then attributed to a smooth transition, on the strength of a result that turned out to come from a bug in our own code that manufactured the very artefact we were trying to rule out. A repeatability test that reported zero variance because the random seed was being overwritten. An aggregate result that was two channels cancelling. And a temperature fix that nearly halved the error in simulation and failed on the instrument.

Every one was caught the same way, and never by thinking harder:

- Build the version of the measurement that the boring explanation would produce, and compare against it. Almost every real finding in Sections 3.3.2 and 3.3.3 of the report came from a control, and almost every error came from skipping one.
- Repeat with different random starting points before believing a difference. Several apparently clear results did not survive this, and one apparently perfect result - zero variance across three runs - turned out to mean the experiment never ran.
- Ask where a proposed mechanism can physically act before asking whether it could produce the observed signal. Two of the six wrong explanations were excludable by this alone, before any code was written.
- Never let a test run entirely in simulation decide a question about the gap between simulation and reality. This arose twice, pointing in opposite directions, and got the answer wrong both times until real-data checks were applied.

Every diagnostic that a conclusion depends on now lives in a numbered script and can be re-run with one command. That was itself a lesson: the measurement underpinning the two-population claim had been made by a script that was never saved, and two records of it in our own notes disagreed with each other. It could not be reproduced, and when we rebuilt it properly with a control, it did not survive.